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  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "09-07-2026_17:43:53",
  "variant" : 2,
  "subjectName" : "mathematics",
  "categories" : [ "Combinatorics", "Planimetry", "Mathematical Logic" ],
  "questions" : [ {
    "question" : "Which principle is also known as the Dirichlet Box Principle?",
    "answers" : [ "Multiplication Principle", "Inclusion-Exclusion Principle", "Pigeonhole Principle",
    "Bijection Principle" ],
    "correctIndex" : 2
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    "question" : "In how many ways can 4 people be seated in a straight line?",
    "answers" : [ "6", "24", "16", "12" ],
    "correctIndex" : 1
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    "question" : "What is the sum of the interior angles of a regular hexagon?",
    "answers" : [ "540 degrees", "720 degrees", "900 degrees", "360 degrees" ],
    "correctIndex" : 1
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    "question" : "According to Brahmagupta's formula, what does it calculate?",
    "answers" : [ "Volume of a pyramid", "Area of a scalene triangle", "Area of a cyclic quadrilateral",
    "Perimeter of an ellipse" ],
    "correctIndex" : 2
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    "question" : "What is the length of the hypotenuse of a right-angled triangle with sides 5 and 12?",
    "answers" : [ "15", "14", "17", "13" ],
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    "question" : "Which theorem states that the square of the hypotenuse is equal to the sum of
    squares of the other two sides?",
    "answers" : [ "Ptolemy's Theorem", "Pythagorean Theorem", "Euclid's Theorem", "Thales's
    Theorem" ],
    "correctIndex" : 1
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    "question" : "If P is true and Q is false, what is the truth value of the implication  $P \rightarrow Q$ ?",
    "answers" : [ "False", "True", "Contingent", "Undefined" ],
    "correctIndex" : 0
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    "question" : "What logic law simplifies 'not (A and B)' into '(not A) or (not B)'?",
    "answers" : [ "Commutative Law", "Distributive Law", "De Morgan's Law", "Associative Law" ],
    "correctIndex" : 2
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}, {  
  "question" : "Which logical connective represents an 'exclusive or' (XOR)?",  
  "answers" : [ "Both inputs are false", "Both inputs are true", "At least one input is true", "Exactly  
one input is true" ],  
  "correctIndex" : 3  
}, {  
  "question" : "What is the negation of the statement 'All primes are odd'?",  
  "answers" : [ "Some primes are odd", "No primes are odd", "There exists at least one prime that is  
not odd", "All primes are even" ],  
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